

Runbook — Confluent Platform + Flink on Minikube

End-to-end operational guide for running `confluent-kafka-isotope` locally on **Minikube**: cluster → Kafka/Confluent Platform → Flink → SQL reports → traffic → observe → teardown. Every step maps to a target in the [Makefile](#), which is the source of truth.

This is the **Confluent Platform (self-managed) on Minikube** path. The Confluent Cloud for Apache Flink (CCAF) path is entirely different — Terraform-driven via `make cc-flink-reports-up`, no Minikube — see [README §4.5](#).

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0. Prerequisites (once per machine)

```
make install-prereqs    # docker, kubectl, minikube, helm, gettext,
gradle, openjdk17
make check-prereqs     # verify they're on PATH
```

Default Minikube sizing (override via env): `MINIKUBE_CPUS=6`, `MINIKUBE_MEM=20480`, `MINIKUBE_DISK=50g`. The node architecture is auto-detected so the right `cp-flink` image (amd64/arm64) is selected.

1. Cluster + Confluent Platform

```
make cp-up              # = check-prereqs → minikube-start → operator-
install → cp-deploy
make cp-watch           # watch pods come up (Ctrl+C to exit); or: make
cp-status
```

`cp-up` boots Minikube, installs the CFK (Confluent for Kubernetes) operator, and deploys Kafka (KRaft) + Schema Registry + Connect + ksqlDB + REST Proxy + Control Center into the `confluent` namespace.

`cp-up` deliberately does **not** bring up Flink — run [step 2](#) separately.

2. Flink

```
make flink-up          # cert-manager → Flink operator → CMF → CMF env →
session cluster
make flink-status      # (~5 min the first time)
```

This installs cert-manager, the Confluent Flink Kubernetes Operator, CMF (Confluent Manager for Apache Flink), creates the `dev-local` Flink environment, and deploys a `cp-flink session cluster` (Flink 2.1.2) with 8 task slots.

CMF license. The `cp-cmf` image ships with a date-locked trial license. Once it expires, the CMF pod `CrashLoopBackOffs` on startup (`LicenseInitiator` throws) and `make flink-up` fails at the CMF readiness wait. To run past expiry, supply a Confluent license via a secret and point `CMF_LICENSE_SECRET` at it:

```
kubectl create secret generic confluent-license-for-cmf -n confluent \
  --from-file=license.txt=/path/to/license.txt # key MUST be
license.txt
make flink-up CMF_LICENSE_SECRET=confluent-license-for-cmf
```

Export `CMF_LICENSE_SECRET` in your shell to make plain `make flink-up` pick it up.

3. Port-forward Kafka

```
make kafka-pf-up      # localhost:30092 → Kafka, localhost:8081 → SR
```

Prereq for everything the host-run gradle app does — `App.java`'s defaults already point at `localhost:30092` / `localhost:8081`. Stop with `make kafka-pf-down`.

4. Deploy the 7 Flink SQL reports

```
make flink-reports-up  # builds PTF shadow JAR → uploads to JobManager →
pre-creates
                        # 7 sink topics → applies source/view/sink DDL →
submits
                        # 7 INSERT INTO streaming jobs (one per report)
```

Five reports are pure Flink SQL; two are JAR-backed `ProcessTableFunctions` (`LatencyPercentilesPTF`, `StuckTracePTF`). Sink topics use Apache Flink's `avro-confluent` format — SR-framed Avro, auto-registered on first write — so Control Center renders the rows natively. Drop everything with `make flink-reports-down`.

5. Drive traffic (required to see report rows)

All report jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, which only emit when the watermark advances past `window_end`. Spread records across **multiple** windows:

```
# 30 records, 5s apart ≈ 2.5 min of event-time → spans 3+ windows
for i in {1..30}; do ./gradlew :app:run --args="place burst-$i" -q; sleep
5; done
```

Or run the pipeline-position stages, each in its own terminal:

```
./gradlew :app:run --args="place hello" # origin → orders.placed
./gradlew :app:run --args="enrich"      # orders.placed →
orders.enriched
./gradlew :app:run --args="fulfill"     # orders.enriched →
orders.fulfilled
./gradlew :app:run --args="ship"        # terminal consume
orders.fulfilled
```

Wait ~90s after the **last** record before checking results — that's the watermark catching up.

`stuck_trace_alerts_1m` only fires for a trace that goes ≥60s of event time without a fresh hop; to exercise it, send one record to `orders.placed`, skip the `enrich/fulfill` hops, and keep sending unrelated records so the watermark crosses `event_time + 60s`.

6. Observe

```
make c3-open          # Control Center – report topics render natively
(Avro+SR)
make flink-sql         # interactive Flink SQL client inside the
JobManager pod
make flink-ui          # Flink job graph / watermarks
make metrics-up        # optional: Prometheus + Grafana for the stateless
meters
```

The metrics showcase has its own runbook — [k8s/monitoring/README.md](#) (and the meter/PromQL reference in [docs/metrics.md](#)). Stop the background port-forwards with `make c3-stop` / `make flink-ui-stop` / `make metrics-down`.

7. Teardown

Pick the depth:

```
make flink-reports-down # drop reports / views / functions only
make metrics-delete     # remove the Prometheus/Grafana showcase (pods +
namespace)
```

```
make flink-down          # Flink cluster + CMF + operator + cert-manager
make cp-down            # CP + operator (keeps Minikube running)
make confluent-teardown # everything + stop Minikube
make nuke                # confluent-teardown + minikube-delete +
uninstall-prereqs (factory reset)
```

Minimal path (no Flink)

To just watch a trace propagate without the Flink SQL reports:

```
make cp-up
make kafka-pf-up
./gradlew :app:run --args="place hello" # then enrich / fulfill / ship
```

Flink ([step 2](#)) and the reports ([step 4](#)) are only needed for the SQL report topics.

Troubleshooting

- **Pods stuck Pending.** Minikube is under-resourced — raise `MINIKUBE_CPUS` / `MINIKUBE_MEM` and `make minikube-delete && make cp-up`.
- **Flink job submission fails (`services forbidden`).** The supplemental RBAC didn't apply — `make flink-rbac`.
- **No report rows.** Almost always the watermark — see [step 5](#); traffic must span multiple 1-minute windows and you must wait ~90s after the last record.
- **App can't reach Kafka.** `make kafka-pf-up` isn't running, or the forward died — re-run it and confirm `localhost:30092` / `localhost:8081` are live.
- **Control Center's Flink tab is blank.** CMF proxy connectivity — `make cmf-proxy-inject` (and `make cmf-proxy-logs` to debug).
- **`make flink-up` times out waiting for the CMF pod / CMF `CrashLoopBackOff`.** Check `kubectl logs -n confluent -l app.kubernetes.io/name=confluent-manager-for-apache-flink`. If you see `Trial license ... expired / LicenseInitiator ... Constructor threw exception`, the image's embedded trial license has expired — supply your own via `CMF_LICENSE_SECRET` (see the CMF license note in [step 2](#)). Wiping the CMF PVC does **not** reset it; the expiry is baked into the image.